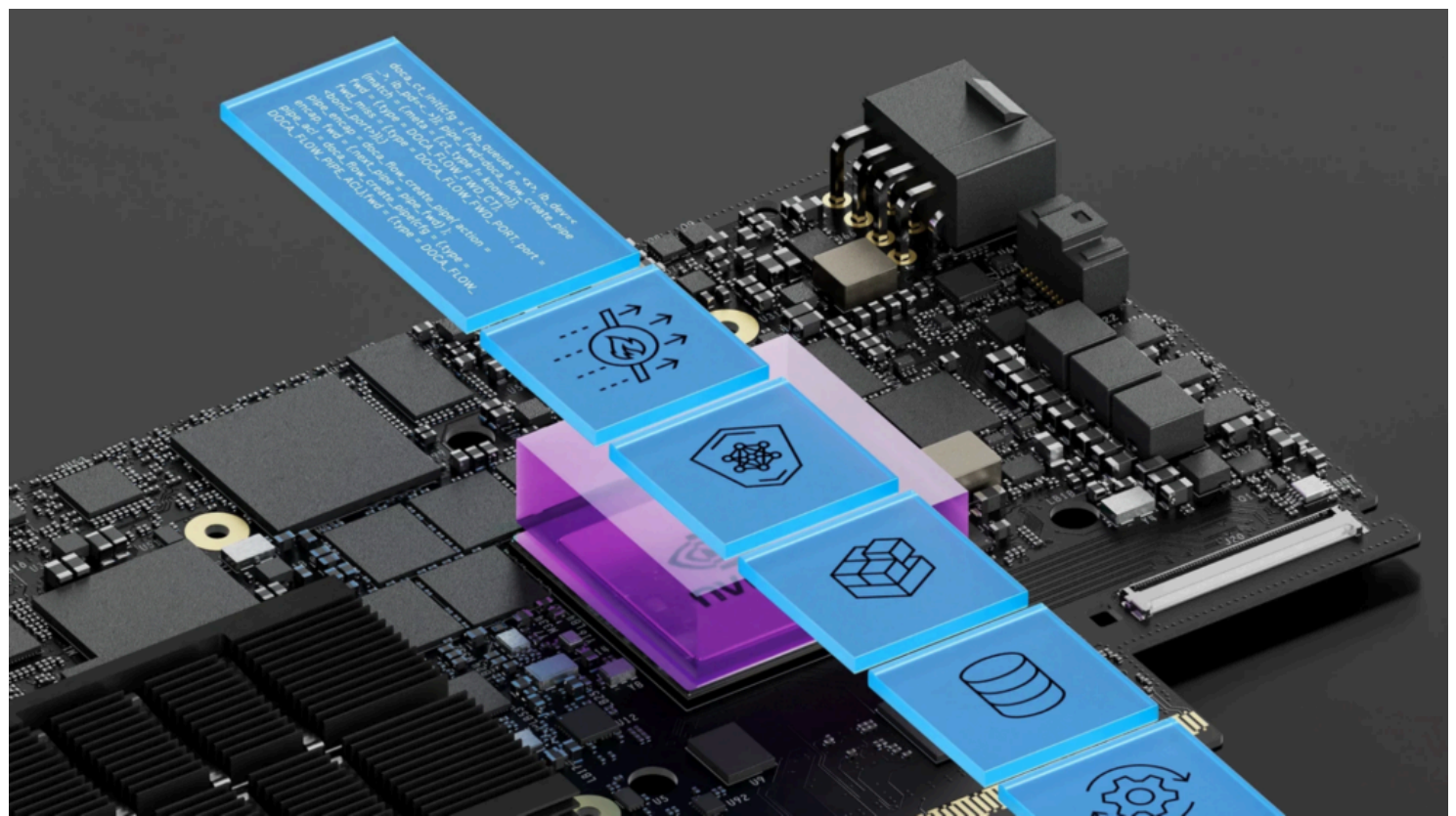




Spotlight: NVIDIA BlueField DPUs Power the VAST Data Platform for AI Workload Optimization



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By [Tim Lustig](#)

As the demand for sophisticated AI capabilities escalates, [VAST Data](#) introduces the VAST Data Platform, now enhanced with [NVIDIA BlueField DPUs](#). This innovation is tailored to meet the



This post presents how BlueField DPUs provide VAST with a significant boost in both performance and efficiency to ensure faster data access and processing speeds, while reducing power consumption and space requirements.

Challenges of managing AI workloads

Optimizing AI workloads requires managing large volumes of unstructured data, ensuring high-speed data access, and maintaining robust data security. Traditional data storage and processing systems often struggle with latency, inefficiency, and scalability issues, which can hinder the performance of AI applications. Additionally, the need for real-time data processing and stringent security adds to the complexity of managing AI workloads effectively.

Benefits of NVIDIA BlueField DPUs

BlueField-3 DPUs empower organizations to meet the demanding requirements of modern AI workloads, ensuring faster data access, robust security, and improved overall efficiency. Read on to learn more about the specific benefits NVIDIA BlueField DPUs bring to the VAST Data Platform and AI workloads.

Improving storage efficiency, data integrity, and security

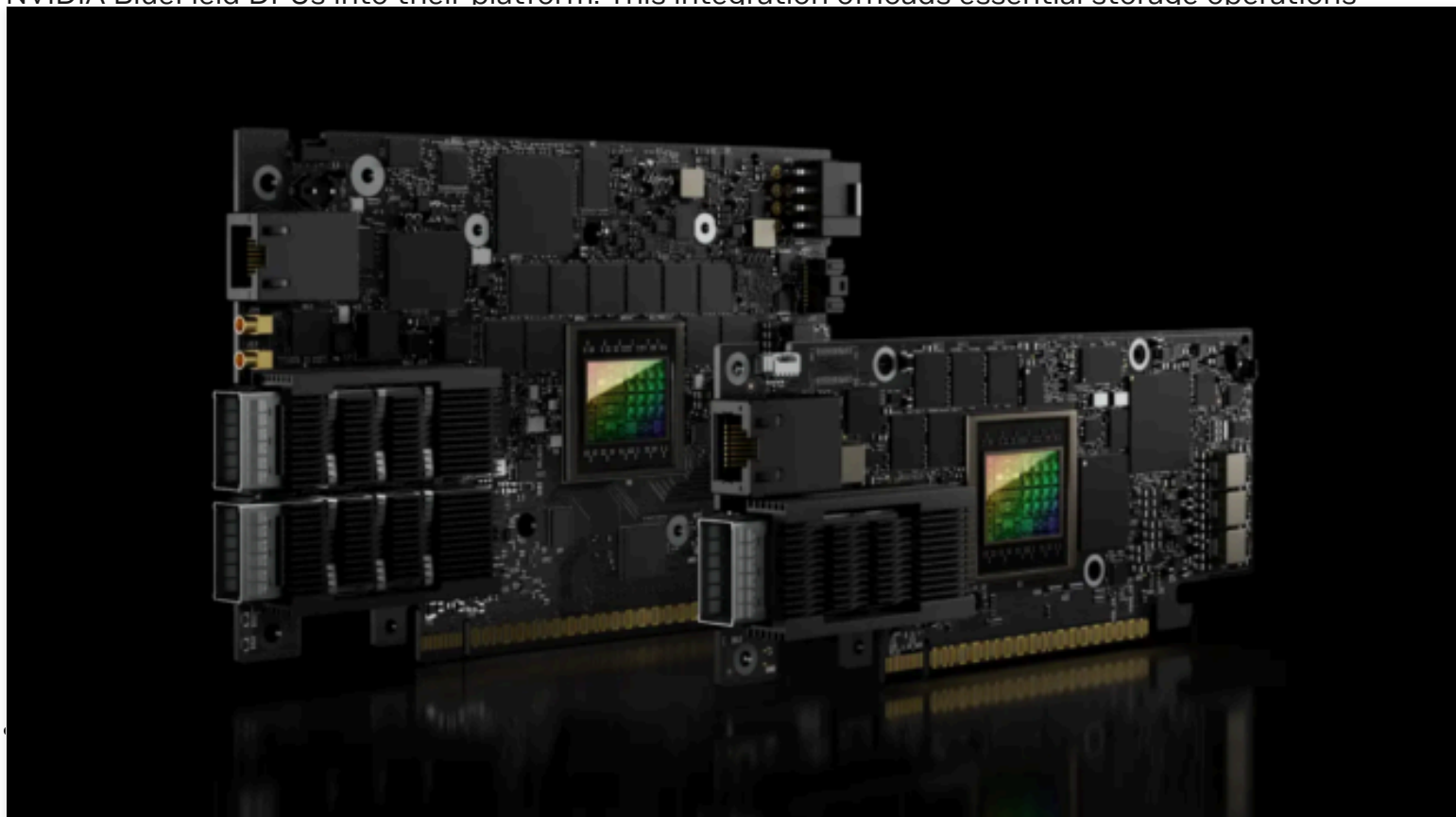
The integration of NVIDIA BlueField DPUs into the VAST Data Platform represents a major leap in storage processing technology. By offloading essential storage operations to the DPU, the results are reduced power consumption and space while enhancing storage networking bandwidth, boosting performance, and ensuring scalability. These enhancements are vital for handling the large data volumes typical in AI applications, facilitating faster data access and processing speeds.

VAST Data's latest offering represents a pivotal development in data platforms for AI environments, combining high-density storage with cutting-edge BlueField DPU technology. This powerful combination ensures exceptional performance, maximizes efficiency, and provides the scalability needed for the most demanding AI environments.



Figure 1. NVIDIA BlueField DPUs enhance storage performance, security, and efficiency in the server, storage controller, and storage enclosure

In VAST Data's traditional architecture, CNodes (compute nodes) are x86 servers responsible for running storage protocols and management services. VAST's unique approach involves integrating NVIDIA BlueField DPUs into their platform. This integration offloads essential storage operations



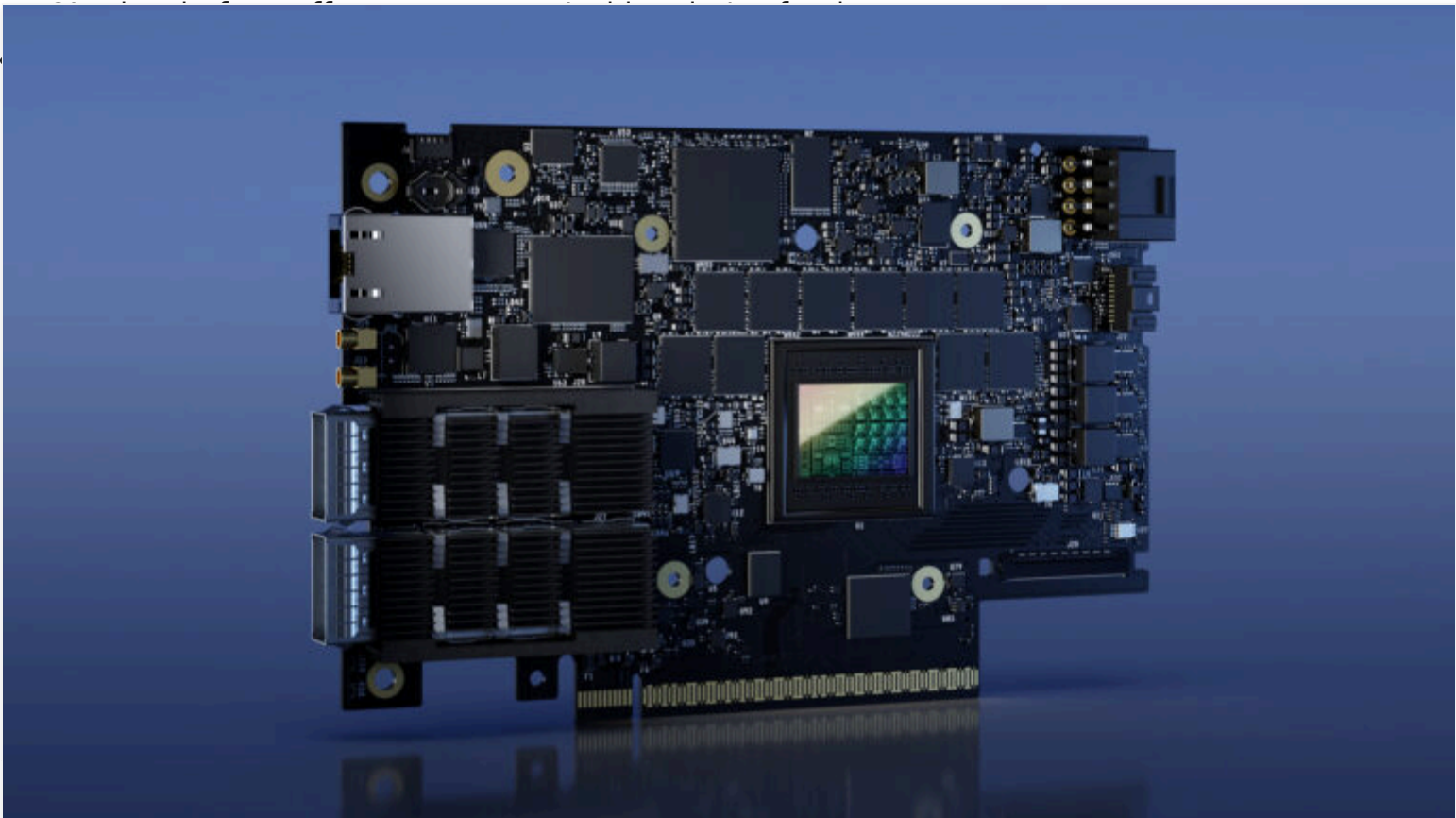
Supermicro Launches NVIDIA BlueField-Powered JBOF to Optimize AI Storage

- **Quality of service:** Each GPU server is equipped with a dedicated BlueField-3 DPU to power the VAST parallel services operating system. This allows each DPU to read and write into the shared



Accelerate AI Infrastructure Using an NVIDIA BlueField-3 DPU Integration with DDN Storage

- **Improved efficiency:** By reducing power consumption by 77% and rack space requirements by



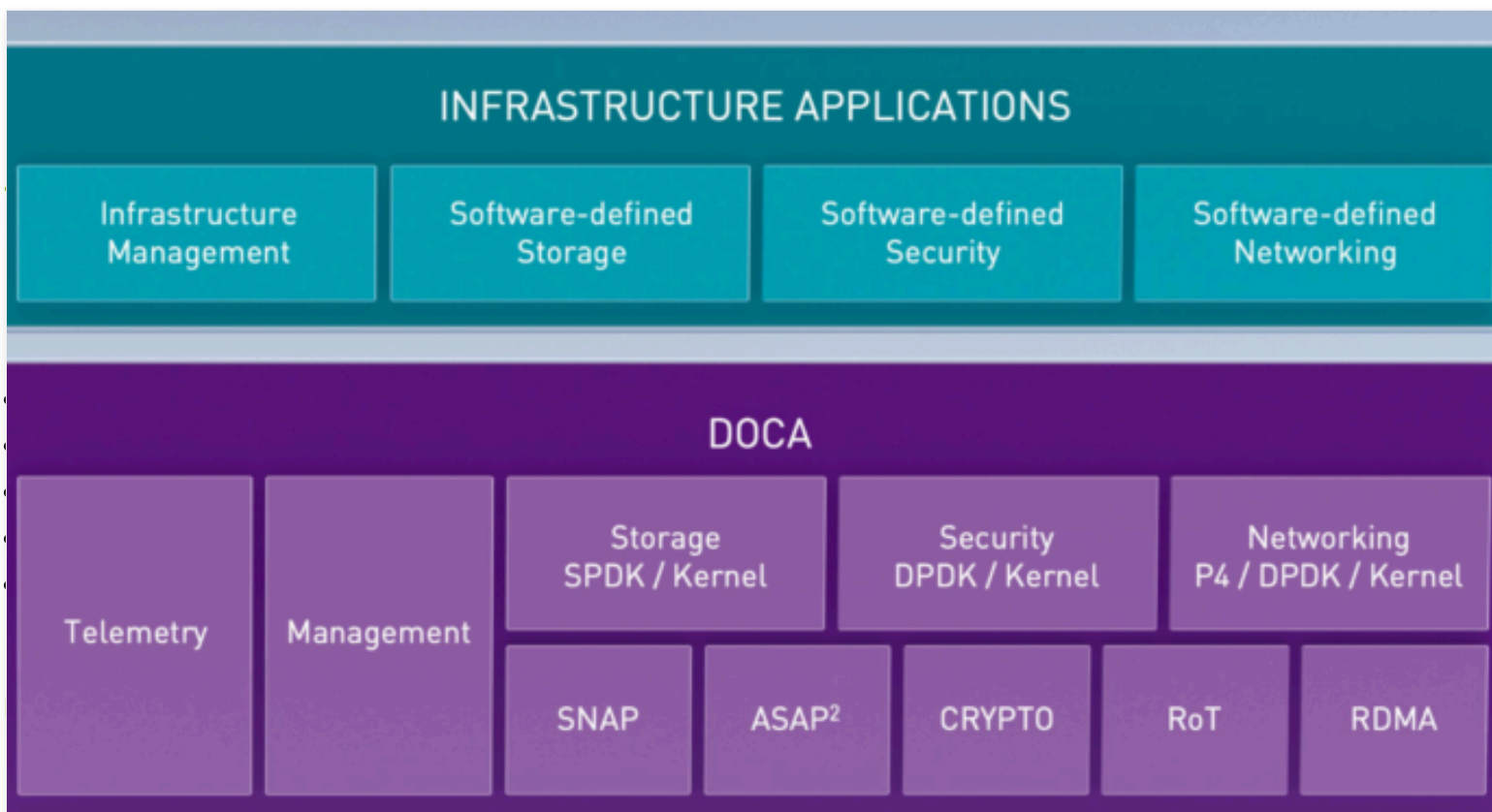
Power the Next Wave of Applications with NVIDIA BlueField-3 DPUs

Data has successfully enhanced its AI cloud architecture, delivering unprecedented performance, security, and efficiency. This integration enables VAST Data to offload critical networking, storage,



Transform the Data Center for the AI Era with NVIDIA DPUs and NVIDIA DOCA

architectures.



Programming the Entire Data Center Infrastructure with the NVIDIA DOCA SDK



ABOUT THE AUTHORS

About Tim Lustig

Tim Lustig is the director of Corporate Ethernet Marketing at NVIDIA, where his work includes the development of collateral and content creation for sales enablement and product launches. He also develops product promotional activities and outbound communication plans to raise brand awareness for the NVIDIA industry-leading Ethernet product lines.

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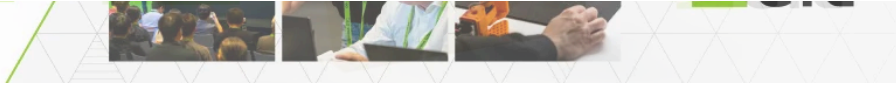


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